Improving interactivity and consistency of the Rzk Language Server through on-edit reevaluation.

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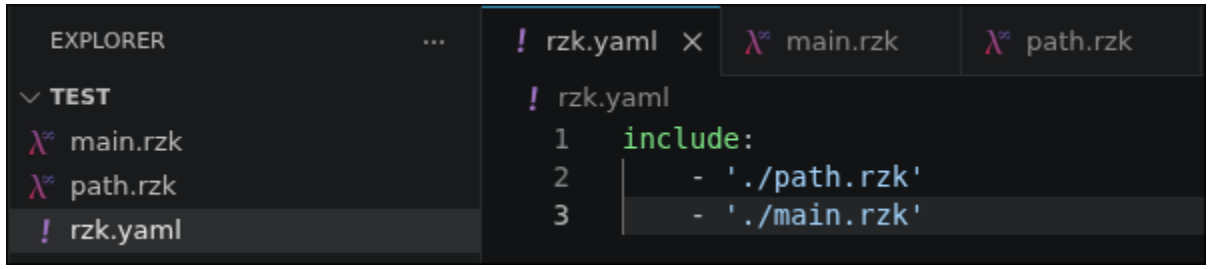
Rzk

Rzk is a proof assistant based on Riehl and Shulman's type theory [1] with shapes. It is used in some leading math formalization works:

- Formalizing the Infinite-Categorical Yoneda Lemma [2]
- Simplicial HoTT and synthetic infinite-categories [3]

Rzk projects are structured around a main 'rzk.yaml' file, which defines a flat list of included files, alongside all other '.rzk' files containing the actual declarations.



State of Rzk LSP server

- Aware of unsaved changes in IDE, but mostly ignores them and read from disk
- Stores unsaved IDE changes (LSP's `document/didChange` call) in virtual file system (VFS)
- Half of features are buffer-aware (semantic highlighting, formatting, go-to-definition), but are triggered only upon saving 'rzk.yaml'
- The typechecker always reads from the disk
- Typecheck is incremental on the level of separate modules, and its results are cached
- On the level of whole project change in an early module invalidate the entire remainder of 'rzk.yaml'

Problem statement

Goal: Improve project-level cache invalidation, make typechecker aware of unsaved changes in IDE and add on-edit evaluation.
Evaluation criteria:

- Typechecking is correct if its result for buffered changes are the same as for 'cold' typechecking from disk.
- Latency for typechecking on large projects should be significantly lower for changes in early modules of 'rzk.yaml' and nearly same for later ones, compared to current Rzk version.

State of the Art: `agda --lsp`

- Agda is a dependently typed programming language and an extension of Martin-Löf's type theory.
- File-wise: no caching. On request, agda reevaluates the whole file.
- Project-wise: compiled module cache.
- Agda does not evaluate on edit, only on save or by explicit request.

State of the Art: `Lean.Server`

- Lean is a functional programming language and theorem prover built for formalizing mathematics and for formal verification.
- File-wise: directed acyclic graph (DAG). Invalidates only changed declaration and its dependencies.
- Project-wise: compiled module cache.
  Only open files and files imported by them are evaluated.
  Invalidation is implemented through a file dependency graph.
- On edit: immediate restart on each keystroke.

State of the Art: `coq-lsp`

- `coq-lsp` is a language server for Coq (Roqc) theorem prover.
- File-wise: linear cascade invalidation. It invalidates rest of the file below changed region (including edited declaration).
- Project-wise: compiled module cache.
  Invalidation is implemented by monitoring compiled `.vo` files for each module; when a file changes, all corresponding cache is cleared.
- On edit: debouncing and then lazy evaluation, with priority on the current cursor position.

Plan of work

1. Make typechecking aware of unsaved changes
2. Decide on a policy for on-edit project evaluation and implement it
3. Design and implement project-level module dependency graph and cache invalidation
4. Testing and Evaluation

References

1. A type theory for synthetic ∞-categories
2. Formalizing the Infinite-Categorical Yoneda Lemma
3. Simplicial HoTT and synthetic ∞-categories